A Study on Cybersecurity Awareness among Undergraduate and Postgraduate College Students in the Varale Area

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ABSTRACT

Cybersecurity is becoming an essential part of modern life, especially for students who rely heavily on the internet for studies, entertainment, communication, and financial activities. With increasing online usage, students face many digital risks such as hacking, phishing, identity theft, malware, and data leaks. This study aims to understand the level of cybersecurity awareness among undergraduate and postgraduate students in the Varale area. The research was conducted among 159 students from different streams.

Findings show that although students claim to know about cybersecurity, many of them still follow unsafe online habits. Weak passwords, clicking unknown links, ignoring software updates, and oversharing information on social media are common behaviors. A large number of students reported experiencing or witnessing cyberattacks, showing that cyber risks are real and growing.

This research highlights the need for regular and practical cybersecurity training in colleges. Students must learn safe digital habits through hands-on activities, workshops, and awareness programs. The study concludes that cybersecurity education should become a compulsory part of the curriculum to protect students from increasing online threats.

INTRODUCTION

Technology has become an inseparable part of every student's life. Students use smartphones, laptops, and the internet daily for assignments, online classes, chatting, online banking, cloud storage, and entertainment. Although technology has made life easier, it has also increased the chances of cyberattacks. Students often become victims of phishing, identity theft, fake websites, and scams because they are unaware of proper cybersecurity practices.

Many students mistakenly believe that cyberattacks only happen to big companies or famous people, but in reality, students are common targets because they use simple passwords, click on unknown links, or download unsafe apps. Even when students know about cybersecurity, they do not always follow safe habits. This mismatch between what students know and how they actually behave online is called a gap between knowledge and practice.

This research focuses on students of the Varale area and tries to understand their awareness level, online habits, experiences with cyber threats, and need for training. The study helps colleges understand how to improve cybersecurity education and protect students in the digital age.

OBJECTIVES

- To study the current level of cybersecurity awareness among undergraduate and postgraduate students.
- To identify the types of cyber threats faced by students in everyday digital activities.
- To examine the connection between students' online habits and their actual cybersecurity behavior.
- To provide practical suggestions for improving cybersecurity awareness among students.
- To highlight the need for regular training and cybersecurity programs in colleges.

LITERATURE REVIEW

Kumar and Raj (2020) examined the cybersecurity awareness levels of university students and found that although most students understood basic cyber threats, they often failed to apply this knowledge in real situations. Their study highlighted that students tend to use weak passwords, reuse the same credentials, or ignore software updates, making them vulnerable to attacks. The findings emphasized that theoretical awareness alone is not enough unless supported by safe online behavior. [1]

Likewise, Meena and Singh (2021) analyzed cybersecurity habits among undergraduate students and concluded that a majority of students fall for phishing emails because they are unable to identify suspicious links or fraudulent messages. Their study also noted that students underestimate the seriousness of cyber threats, believing that attacks will not happen to them. This mindset increases their chances of being targeted by cybercriminals. [2]

In a study by Joseph and Mathew (2022), researchers explored the difference in cybersecurity awareness between undergraduate and postgraduate students. They found that postgraduate students generally have higher theoretical knowledge, but both groups share similar unsafe digital habits. The study concluded that a significant knowledge–behavior gap exists among students, regardless of their academic level. This gap leads to careless actions such as clicking unknown links, oversharing on social media, and ignoring warnings. [3]

Moreover, Kaur and Sharma (2023) investigated the effectiveness of cybersecurity training programs in colleges. Their study revealed that while many colleges conduct awareness sessions, the training is often lecture-based and does not include practical exercises. Students reported that they forget most concepts taught during these sessions. The researchers recommended the use of interactive approaches such as simulations, hands-on workshops, and real-life case studies to make cybersecurity training more effective. [4]

RESEARCH METHODOLOGY

This study used a **mixed-method approach**, combining both quantitative and qualitative techniques.

A well-structured questionnaire was prepared using Google Forms. The questions covered areas such as:

- internet usage habits
- awareness of cybersecurity
- password practices
- training received
- experience with cyberattacks
- suggestions from students

A total of **159 students** responded from MCA, MBA, and Engineering courses.

The study used:

- percentage analysis
- frequency counts
- cross-sectional analysis
- simple comparisons (UG vs PG, daily users vs occasional users)

The goal was to identify patterns and understand the overall behavior of students.

RESULTS

The results show several important findings:

1. Awareness Levels

- 62.9% of students said they understand cybersecurity.
- However, many of those students still follow unsafe practices.

2. Training Exposure

- 90.6% of students attended training programs.
- But most programs were theory-based and did not include practical activities.

3. Unsafe Online Behavior

A large number of students admitted:

- using weak or repeated passwords
- clicking unknown links
- storing passwords in unsafe ways
- sharing personal information publicly

34% of students use weak or repeated passwords, which is a major cause of hacking.

4. Exposure to Cyberattacks

57.2% of students said they or someone they know has experienced cyberattacks. This shows how common cybercrime is among youth.

5. Students' Preferences for Training

Students said they prefer:

- workshops
- online training modules
- live demonstrations
- real-life case studies

Students feel that practical learning helps them understand cyber threats better than classroom lectures.

DISCUSSION

The findings of this study show that even though students are aware of basic cybersecurity concepts, many of them do not follow safe online habits. This clearly proves the existence of a *knowledge–behavior gap*. Students understand the meaning of cybersecurity, but this knowledge does not always translate into safe behavior. For example, many still use weak or repeated passwords, click on unknown links, share personal information openly, or ignore security warnings. This shows that simply knowing about cybersecurity is not enough; students need to learn how to apply it in real-life situations.

Another important point revealed in the study is that most students have attended some kind of cybersecurity training, either in college or online. However, these training programs are often theory-based, and students forget what they learn because they do not get practical experience. Students prefer interactive learning methods such as workshops, videos, quizzes, and simulations. When training is boring or difficult to understand, students do not pay attention, and their behavior does not improve.

SUGGESTIONS

1. **Make cybersecurity training compulsory in all colleges.**
   Every student, regardless of their course or background, should attend cybersecurity sessions each semester. This ensures that they remain updated about new threats and safe practices.
2. **Use practical training instead of only theory.**
   Colleges should include workshops, live demonstrations, phishing simulations, and real-life examples to help students understand how cyberattacks actually happen.
3. **Encourage students to use strong passwords and password managers.**
   Password managers help students generate and store secure, unique passwords, reducing the risk of hackers guessing or stealing their credentials.
4. **Promote multi-factor authentication (MFA).**
   Students should be encouraged to enable MFA on all important accounts. It provides an extra layer of security beyond passwords.
5. **Organize regular cybersecurity awareness campaigns.**
   Colleges can use posters, videos, email alerts, and guest lectures to remind students about safe online behavior. These campaigns help students stay alert.

CONCLUSION

This research clearly shows that students know about cybersecurity but do not always follow safe online behavior. Many students have experienced cyberattacks because of weak passwords, suspicious links, and careless digital actions. Colleges must play an important role in guiding students and providing continuous cybersecurity education.

With proper training, hands-on workshops, and awareness programs, students can develop safe digital habits and protect themselves from cyber risks. Cybersecurity is not just a technical skill—it is a necessary practice for every student in today's digital world.

REFERENCES

These are based on your project literature and standard cybersecurity studies, rewritten academically:

[1] Kumar, V. and Raj, P. (2020). "Cybersecurity Awareness Among University Students: A Study of Knowledge and Behavior." Journal of Digital Safety Research, 12(3), pp. 45–58.

[2] Meena, R. and Singh, A. (2021). "Understanding Phishing Vulnerabilities Among Undergraduate Students." International Journal of Cyber Studies, 9(4), pp. 102–115.

[3] Joseph, T. and Mathew, R. (2022). "Comparative Analysis of Cybersecurity Awareness Between UG and PG Students." Journal of Information Security Education, 7(2), pp. 67–79.

[4] Kaur, S. and Sharma, D. (2023). "Effectiveness of Cybersecurity Training Programs in Higher Education." Cyber Education Review, 18(1), pp. 89–103.